Guidelines for Reporting About Network Data (GRAND)

The Guidelines for Reporting About Network Data (GRAND) are designed to facilitate clarity and transparency in network research reported in scientific documents (e.g., articles, monographs) by providing researchers, editors, and peer reviewers with guidance about the information that should always be reported. In practice, it will often be necessary to report additional information not included in these guidelines depending on the research question and other contextual details.

Please cite GRAND as: Neal, Z. P., Carley, K., Contractor, N., De Domenico, M., Hollway, J., Labianca, G., Landry, N., Lazer, D., Lubbers, M., Neal, J. W., Ortiz, F., Peixoto, T., Pescosolido, B., Prell, C., Teixeira, A. S., and the GRAND Consortium. (2026). Guidelines for Reporting About Network Data. *Pre-print*.

Frequently Asked Questions

What if a piece of information is not relevant?
The items in GRAND were selected by the GRAND Working Group and Consortium to be relevant in nearly all cases and researchers are encouraged to report all GRAND items. However, GRAND provides guidelines, not requirements.

Some of this information is already provided in another document. Do I need to report it again?
Yes. GRAND ensures that each document reporting on a network contains at least a minimal description of the data. If the data are described in more detail elsewhere, you may cite it and refer readers there for more information.

Where should I report these details?
GRAND should be used to describe the network data in the main text of the document, as opposed to in an appendix or supplementary materials. However, GRAND does not specify how or where these details should be reported. They can appear in any order throughout the document, or can be summarized in a single paragraph or table.

Should I report about raw or analytic data?
In general, GRAND should be used to describe the final network(s) upon which analyses are performed. However, the items in section 4 describe any transformations that were performed on raw data to obtain the final analytic network.

What if my network does not fit these guidelines?
GRAND is designed to be applicable to many different types of networks, but there are limits. For example, GRAND does not currently provide guidance for continuous time or relational event networks, ego networks with only ego-alter ties, or simplicial complexes. In such cases, consider adapting GRAND to your specific case, or follow the spirit of GRAND by ensuring that your document contains enough details for readers to fully understand the key features of your network.

Can I use different terminology?
Yes. GRAND is agnostic about terminology, and recognizes that the same idea may have different names in different fields. The notes often identify potential synonyms, but are not exhaustive. When reporting the details recommended by GRAND, use the terminology that is most appropriate for your audience.

How do I report about multiple networks in the same document?
If your document reports on a multi-wave longitudinal network (e.g., animal interactions in January, February, and March), a set of ego networks (e.g., the personal social networks of Tom, Dick, and Harry), or a set of networks all representing the same kind of system (e.g., communication network in companies A, B, and C), then you can describe these as a single network dataset (see 1.2). However, if your document reports on a set of unrelated networks (e.g., validation of an algorithm using a human social network, an animal interaction network, a transportation network, and a neural network), each network should be described separately using GRAND. In this case, it can be appropriate to describe the networks in an appendix or supplementary information.

Do I need to share my data? Do I need to use GRAND if I've already shared my data?
Sharing network data is a good practice, but it is not part of GRAND. Instead, GRAND is focused on clearly describing network data, regardless of whether that data has been or will be shared.

How do I report all this information in a document with a strict word limit?
The suggested information is probably already in your document, so GRAND simply serves as a helpful reminder. Additionally, the information can often be reported very compactly. For example, multiple items (1.1, 1.4, 1.5, 1.7, 2.1, 2.2, 3.1, 3.2, 3.3, 3.7) can be reported in a single sentence: "The Zachary (1977) Karate Club network was collected between 1970 and 1972 via observation in at American university and contains 34 nodes that represent people, which are connected by 78 unweighted and undirected edges that represent friendship."

1. Data: The items in this section are intended to provide a description of the network dataset as a whole.

#	Item	Reporting guideline	Notes
1.1	**Data**	**If the data are empirical**, report the method originally used to collect them. **If the data are synthetic**, report the data generating model and the values of its parameters.	Empirical data are collected directly (e.g. survey, experiment, observation) or indirectly (e.g., archival, trace; adams, 2020). Synthetic data are generated by a model or simulation (e.g., Preferential attachment model, Exponential Random Graph model).
1.2	**Ethics**	**If the collection or use of these data could present ethical concerns**, report whether institutional or ethical approval was obtained, the approving entity, and approval date.	Ethical concerns can arise when data are collected from or about 'human subjects' (Tubaro et al., 2021), but they can also arise in non-human subjects data that could be used to cause harm.
1.3	**Multiple**	**If the data contain a panel of networks representing the same nodes at different times**, report the number of waves and the interval(s) between waves. **If the data contain multiple networks that each represent the same type of system**, report the number of networks.	Longitudinal panel network data captures the same system at multiple cross-sectional waves (e.g., animal interactions observed at 3 monthly intervals), and can be analyzed using specialized models (e.g., stochastic actor oriented models; Duxbury, 2023). A single dataset can contain a network from each of multiple sites (e.g., communication networks in 10 companies), or from each of multiple egos (e.g., personal social networks of 10 people; Perry et al., 2018).
1.4	**Location**	**If empirical (see 1.1)**, report the location that the data are intended to represent, at a substantively and ethically appropriate level of resolution.	Examples of possible locations include institutional locations (e.g., a company, a school), geographic locations (e.g., a country, a city), and digital locations (e.g., a social media platform).
1.5	**Date**	**If empirical (see 1.1)**, report the date(s) that the data are intended to represent, at a substantively and ethically appropriate level of resolution.	This is not necessarily the same as when the data were collected or compiled (e.g., in 1993 Padgett & Ansell compiled marriage records that represented relationships in 1400-1434).
1.6	**Availability**	Report the URL or DOI linking to a repository where the data or data-generating code are available, or explain why the data are not available in a repository.	This item follows the recommendation that "researchers should share the network data and materials necessary to reproduce reported results...when an associated manuscript is published" (Neal et al., 2024).
1.7	**Citation**	**If the data have an independent citation**, provide a citation to the data.	This item follows the Joint Declaration of Data Citation Principles (Martone et al., 2014). The cited item may provide further details about the data, or may simply give credit to those who originally collected or generated the data.

2. Nodes: The items in this section are intended to describe the nodes in the network(s).

#	Item	Reporting guideline	Notes
2.1	**Nodes**	Report what the nodes are intended to represent. **If the nodes represent more than one type of entity,** describe the network as N-mode and report what each type of node is intended to represent.	Nodes are sometimes also called vertices or actors. The nodes may all represent the same type of entity (e.g., a social network were all nodes represent "people"), or multiple types of entities (e.g., a two-mode pollination network where some represent "plants", and others represent "pollinators").
2.2	**Size**	Report the number of nodes. **If multiple (see 1.3)**, report the number of nodes in each wave/network if practical; otherwise report the mean, standard deviation, and range. **If multimode (see 2.1)**, report this item separately for each type of node.	This value is sometimes called 'network size' or 'graph order', and abbreviated $\|V\|$.
2.3	**Boundary**	**If empirical (see 1.1)**, report how the set of nodes that should be included in the network was defined.	The challenge of determining which nodes *should* be included is distinct from reporting which nodes *are* present, and is known as the 'boundary specification problem' (Laumann, et al., 1983). An ego network defines the nodes that should be included by ego's response to name generator questions. A bounded network defines the nodes that should be included using an *a priori* list of nodes in the system or setting (e.g., a roster). A non-bounded network simply includes any nodes for which data could be obtained (e.g., any accounts included in a social media platform's data release), or which were selected via a sampling procedure (e.g., snowball).
2.4	**Missing**	**If nodes that should be included in the network are missing from the network (see 2.3)**, report the percent or number of missing nodes, or the percent or number of present nodes (i.e., the response rate). **If multiple (see 1.3)**, report this value for each wave/network if practical; otherwise report the mean, standard deviation, and range.	Missingness occurs in a bounded network when nodes that should be included in the network are not included. Laumann, et al. (1983) cautioned that "because it distorts the overall configuration of actors in a system, [it] may render an entire analysis meaningless".
2.5	**Degree**	**If nodes' minimum or maximum degree is constrained by the data collection or generation design**, report any restriction(s) on nodes' degree.	Degree constraints can arise by data collection (e.g., name up to 5 friends; Neal, 2024) or model specification (e.g., in the Barabási & Albert (1999) model, m defines nodes' minimum degree).

3. Edges: The items in this section are intended to describe the edges in the network(s).

#	Item	Reporting guideline	Notes
3.1	**Edges**	Report what the edges are intended to represent. **If the edges represent more than one type of relationship**, describe the network as multiplex or multilayer, and report what each type of edge is intended to represent.	Edges are sometimes also called links, ties, arcs, or connections. The edges may all represent the same type of relationship (e.g., a social network where all edges represent "friendship"), or different types of relationships (e.g., a multiplex/multilayer network where some edges represent "friendship" and other edges represent "kinship".)
3.2	**Direction**	Report whether the edges are directed or undirected. **If directed**, report what the direction of the edge means. **If multiplex (see 3.1)**, report this item separately for each type of edge.	An undirected edge connects two nodes (e.g., i and j are connected, $i - j$), and a network composed of only undirected edges is sometimes called symmetric. A directed edge points from one node toward another (e.g., i sends to j, $i \to j$), and a network that contains directed edges is sometimes called asymmetric. If data were symmetrized, report that the network is undirected, but also see item 4.1.
3.3	**Weight**	Report whether the edges are weighted, unweighted, or signed. **If weighted**, report the meaning, scale of measurement, and mean, standard deviation, and range of weights. **If signed**, report the meaning of positive edges and the meaning of negative edges **If multiplex (see 3.1)**, report this item separately for each type of edge.	In an unweighted network, edges are either present (e.g., 'friends') or absent (e.g. 'not friends'). In a weighted or valued network, present edges are assigned a value that may represent a continuous or ordinal count (e.g., years of friendship), frequency (e.g., of communication), intensity (e.g., of interpersonal closeness), etc. In a signed network, present edges are either positive (.g., 'friendship') or negative (e.g., 'antagonism'). If data were dichotomized, report that the network is unweighted, but also see item 4.2.
3.4	**Selfloops**	**If the network contains self-loops**, report that the network contains self-loops and what they represent.	A self-loop is an edge that connects a node to itself.
3.5	**Bipartite**	**If multimode (see 2.1),** report where edges can be located.	Given a multimode network (e.g. authors and papers), this item distinguishes a bipartite network (e.g., authors can only be connected to papers) from a multilevel network (e.g., authors can be connected to papers or to other authors).
3.6	**Hypergraph**	**If the network is a hypergraph**, report that the network is a hypergraph.	In a hypergraph, a single edge can connect more than two nodes (e.g., co-authorship hypergraph where all authors of an article can be connected via a single edge).
3.7	**Number**	Report the number of edges present. **If multiple (see 1.3)**, report the number of edges in each wave/network if practical; otherwise report the mean, standard deviation, and range. **If multiplex (see 3.1) or signed (see 3.3)**, report this item separately for each type of edge. **If hypergraph (see 3.6)**, report the number of (hyper)edges, and the mean and standard deviation of the size of the hyperedges.	This value is sometimes called 'graph size' and abbreviated $\|E\|$. When a network contains directed edges, each directed edge is counted separately (e.g., a reciprocated dyad $i \rightleftarrows j$ contains *two* edges). When a network contains weighted edges, each present edge is counted, regardless of its weight. In a hypergraph, the 'size' of a hyperedge is the number of nodes that it connects.

4. Transformations: The items in this section are intended to describe transformations that were performed on raw data to obtain the final analytic data.

#	Item	Reporting guideline	Notes
4.1	**Symmetrized**	**If directed edges in raw data were transformed into undirected edges in the final analytic network**, report the transformation method and the percent of connected dyads that had non-reciprocal edges before the transformation.	Examples of symmetrizing transformations include confirmed edges (i.e., transforming only reciprocated edges $i \rightleftarrows j$ into undirected edged $i - j$) and unconfirmed edges (i.e., transforming any directed edges $i \rightarrow j$ and $i \leftarrow j$ into undirected edges $i - j$).
4.2	**Dichotomized**	**If weighted edges in raw data were transformed into unweighted edges in the final analytic network**, report the transformation method and the number or percent of edges that were deleted as a result of the transformation.	Examples of dichotomizing transformations include thresholds (i.e., count all edges with weight $> X$ as present) and backbone models (i.e., count edges with statistically significant weights as present; Neal, 2022).
4.3	**Projected**	**If the final analytic network was generated from two-mode or non-network data such that edges represent similarities, co-occurrences, or statistical associations**, report the transformation method.	Examples of such transformations include two-mode projection (e.g., two people are connected if they attended the same event Breiger, 1974) and (weighted) correlation networks (e.g., two variables are connected to the extent that they are correlated in a sample; Langfelder & Horvath, 2008).
4.4	**Excluded**	**If any nodes or edges that were present in raw data have been excluded from the final analytic network**, report the exclusion criteria and number or percent of nodes or edges that were excluded.	Nodes might be excluded for a variety of both structural (e.g., they are isolates with degree = 0, they are members of a small component) and non-structural (e.g., uncertainty, implausible responses) reasons.
4.5	**Aggregated**	**If any nodes or edges that were separate in raw data have been aggregated in the final analytic network**, report the aggregation method.	Individual nodes are sometimes aggregated into "supernodes" when they are structurally equivalent or belong to the same community. In multiplex networks, edges connecting the same dyad (e.g., i and j are connected by friendship and kinship) are sometimes aggregated into a single edge.
4.6	**Imputed**	**If any nodes or edges were unmeasured in raw data, and their presence or absence was imputed or inferred in the final analytic network**, report the imputation method and the number or percent of nodes or edges that were imputed.	A range of methods exist to impute missing data in networks (Krause et al., 2020), and link prediction methods are sometimes used to infer missing edges.